

More SQL

MBUA 512

Class 4 – More SQL

SQL = Structured Query Language, "es-queue-el" or "sequel". With Michael Winikoff.

- SELECT calculations, ORDER BY (DESC), LIMIT n, named columns, DISTINCT
- WHERE conditions
- GROUP BY, HAVING, and aggregates (min, max, avg, count, sum)

How to write SQL queries & tips

Based on slides by Professor Markus Luczak-Roesch.

Recap: SQL commands

- `CREATE TABLE` and `ALTER TABLE` — define the structure of a table (its *schema*)
 - `INSERT INTO` , `DELETE FROM` , `UPDATE` — change the data in the table
 - `SELECT` — query data
- (there are a lot more than these)

Important: make sure you select `MySQL` (not `Trino`) and your own workspace
(`user_[username]`)

SELECT calculations, ORDER BY, LIMIT

SELECT can include calculations:

```
SELECT concat(firstName, ' ', lastName),  
       floor(datediff(current_date, birthdate)/365)  
FROM people WHERE birthdate IS NOT NULL  
ORDER BY birthdate -- add DESC for youngest  
LIMIT 10;
```

Explanation: `concat` joins strings,
`current_date` is the current date,
`datediff` gives the difference between two
dates in days, `floor` rounds a number down.
`EXTRACT(MONTH from d)` gets the month
part of a date (not shown).

	CONCAT(firstName, ' ', las... /* contact joins strings together */ FL...
1	Pincas Sapwell 98
2	Bernette Blance 98
3	Dion Lantiff 98
4	Sancho Puddephatt 96
5	Harman Ramey 95
6	Gun Beardsley 94
7	Tyrus Forte 94
8	Delmar Arrigo 89
9	Edlin Lenard 87
10	Price Guerin 87

SELECT named columns

Can use `AS name` to give a column a name

```
SELECT concat(firstName, ' ', lastName)
SELECT concat(firstName, ' ', lastName) AS Name
```

	CONCAT(firstName, ' ', lastName) ⌵
1	Brice Taffurelli
2	Delmar Arrigo
3	Edy Jeyness

	Name ⌵
1	Brice Taffurelli
2	Delmar Arrigo
3	Edy Jeyness

SELECT DISTINCT

- Just add `DISTINCT` to remove duplicates *rows* from the result
- Example: select all pilots who have scheduled flights

```
select pilotName from flight, pilot where flight.pilotID=pilot.pilotID  
select DISTINCT pilotName from flight, pilot where flight.pilotID=pilot.pilotID
```

	pilotName
1	Pat Smith
2	Yi Zhang
3	Jim Belich
4	Jim Belich
5	Jim Belich
6	Jim Belich

	pilotName
1	Pat Smith
2	Yi Zhang
3	Jim Belich

WHERE conditions

All sorts of things you can test as conditions in the WHERE part.

- IS NULL - is something NULL? (e.g. `email IS NULL`) - also IS NOT NULL.
- IN: `month IN (1,2,3)` - shorthand for `month=1 OR month=2 OR month=3`.
- LIKE: is a string like a pattern, where `%` means "any string of characters" and `_` means any single character (e.g. `email LIKE '%@%.%'`).
- numerical conditions (`<` `>` `=` `>=` `!=`)
- date conditions (`<` `=` between): `d BETWEEN d1 AND d2`
- logical operators (AND, OR, NOT)
- (not) EXISTS (next slide)

EXISTS

- A WHERE condition is applied to each row on its own. But sometimes we need to check *other* rows when deciding whether to include a given row
- Example: find all people who share a birthday: the condition for a row is whether someone *else* (another row) has the same birthday.
- An EXISTS condition allows us to check a condition for a given row against more than just that row.

```
SELECT personID, firstName, lastName, birthdate FROM people p
WHERE EXISTS (select * from people p2 where p.birthdate=p2.birthdate AND p.personID != p2.personID);
```

Note use of names (p1 , p2) to refer to row in the subquery.

UNION, INTERSECT

- These combine the results of queries.
- UNION: include all rows from both queries
- INTERSECT: include only rows that are in both results. Example: all pilots who fly both a Boeing and an Airbus (can't do this with a WHERE!)

```
(SELECT pilotName from flight, pilot, aircraft
WHERE flight.pilotID=pilot.pilotID AND flight.planeModel=aircraft.planeModel
AND planeManufacturer='Airbus')
INTERSECT
(SELECT pilotName from flight, pilot, aircraft
WHERE flight.pilotID=pilot.pilotID AND flight.planeModel=aircraft.planeModel
AND planeManufacturer='Boeing');
```

Note two join conditions, since we are joining 3 tables.

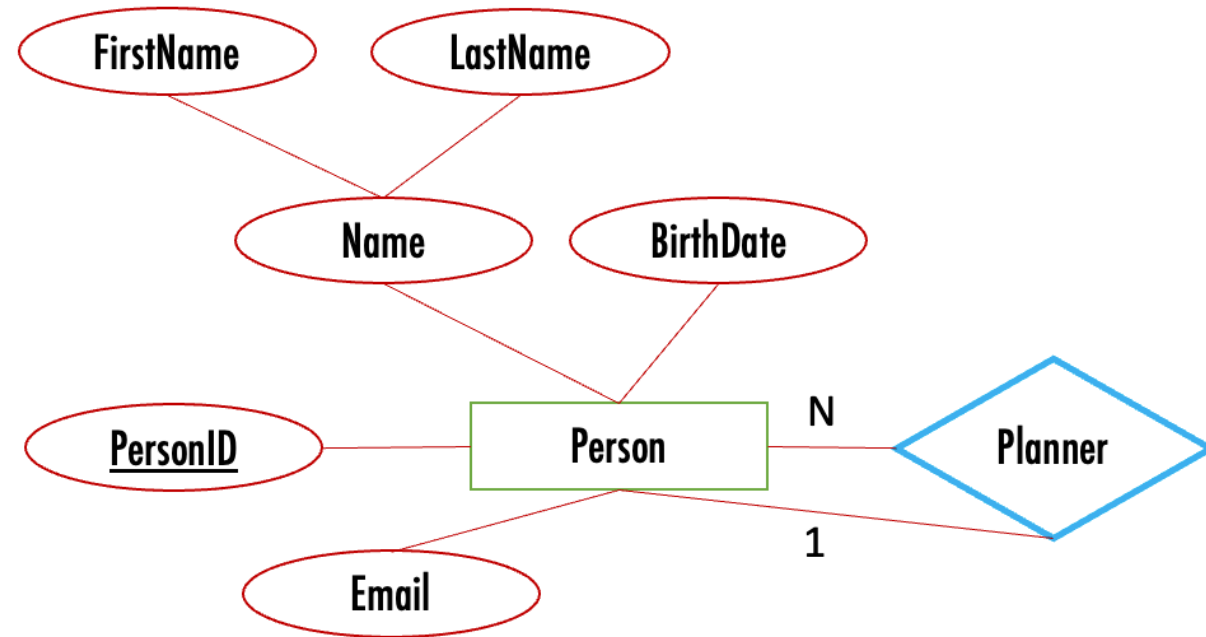
GROUP BY, HAVING, aggregates

Aggregates (min, max, avg, count, sum)

Running Example: birthday

- Scenario: send a message to people on their birthday, and remind their designated party planner to plan a party earlier
- Data: name (first, last), birthdate, email address, and designated party planner

Use [this file](#) to create the database.



Some queries to write (1/2)

1. Find people whose birthday is today so we can email them (combine first and last names)
2. Find people with a special birthday (e.g. 100) coming up (hint: need to calculate age, use FLOOR), order by closest birthday
3. Find people with a birthday in August, September or October so we can email their planner (hint: use IN)

Some queries to write (2/2)

Administrative tasks:

4. Find people with missing or invalid email addresses (hint: use IS NULL AND ... NOT LIKE)
5. Find where two (or more) people have the same email address (hint: use EXISTS)
6. Find people who are not planners for anyone else and do not have a birthday (hint: use EXISTS)

Analysis task:

7. how many people have a birthday in each month? (hint: use GROUP BY and EXTRACT(MONTH from ...))

Tips for writing SQL

- Build it up in pieces, testing each piece (* can be useful)
- Think about:
 - What information you need to see in the result (which columns ...) ... values?
Calculations? Aggregate calculations (e.g. SUM)?
 - What tables they come from, and how you need to JOIN the tables
 - What constraints to use ... and how to express them
 - WHERE if based on row
 - WHERE ... AND EXISTS – if not (or sometimes UNION, INTERSECT)

More tips

- `' '` (single quotes), not `" "` (double quote)
- DISTINCT – the whole *row* needs to be distinct, so sometimes select fewer columns
- GROUP BY ... HAVING ... - the "HAVING" goes just after the "GROUP BY ..."
- NULL is not the same as the empty string
- Too many rows in result? Check JOIN conditions
- No rows in result? Check conditions, and perhaps remove some
- Using EXISTS – make sure there is also an "AS ..." (so can refer to parent query in sub-query)

Activities

Activities – More SQL

1. Go to learn.datascie.nz and login if needed
2. Select MySQL (not Trino) and your own workspace (`user_[username]`)
3. If needed, create the flight DB using [this file](#)

Write queries to answer the following questions

TO COMPLETE

Activity – ...

~ 10 minutes, in groups